# River Flow

In this mini project, we study the river flow and mixing at the junction of the Rhine and Neckar rivers in Mannheim, Germany. You can check the location on Google Maps to determine the flow direction of the rivers. Please note that the geometry used in this project is simplified, and the boundary conditions (river inlet velocities and temperatures) are hypothetical. The purpose of this project is to familiarize you with the various steps involved in your main project.

## 1. Geometry

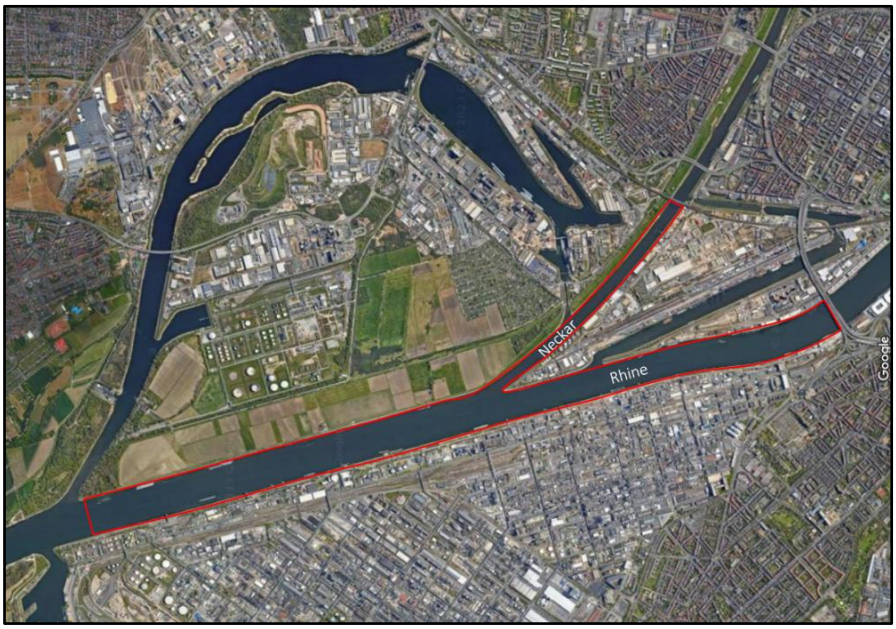



Figure 1: Satellite view of the junction between the Rhine and Neckar rivers in Mannheim, Germany, with the domain of interest outlined in red.

Figure 1 shows a satellite view of the junction between the Rhine and Neckar rivers. The domain of interest for this project is marked by the red lines in the figure. To import the geometry into DesignModeler, the first step is to extract the river boundaries using mapping software. This will provide a set of coordinates defining the river boundaries, which we can use to construct the geometry. The file “Digitized_River.js” contains these coordinates in a format that can be directly imported into DesignModeler as a sketch.

- Open DesignModeler and make sure the units are set to Meter (m).
- Enable Large Model Support by selecting Units, Large Model Support, On.
- Import the “Digitized_River.js” file using File, Run Script.
- As you can see, the geometry consists of three different lines corresponding to the three main solid walls (riverbanks) involved in our geometry. At this stage, the geometry is not fully closed (at the locations marked by the red lines), so a surface cannot yet be created.
- Close the geometry by inserting lines at the locations of the river inlets and outlet.
- Create a surface. Now we are ready for meshing!

## 2. Meshing

In this section, a set of guidelines is provided for generating a suitable mesh for the numerical simulation. The "Basic" subsection outlines the essential steps for the mesh generation process, while the "Recommendations" subsection provides additional tips for achieving a high-quality mesh.

### Basic:

- Under Defaults, make sure the Physics Preference is set to CFD.
- Under Defaults, set the Element Size to 15 m.
- Create the following Named Selections for the boundary:
  - Rhine_in for the inlet of the Rhine.
  - Neckar_in for the inlet of the Neckar.
  - Rhine_out for the Rhine outlet.
  - The three riverbanks (the solid lines in the boundary) as Wall_1, Wall_2, and Wall_3.

### Recommendations:

You can apply different meshing techniques by right-clicking Mesh and selecting Insert:

- Use Edge Sizing for each boundary line. You can also choose to apply a bias to the distribution of mesh elements, allowing areas to have varying cell density.
- Through the Method option, you can specify the meshing method and define the dominant cell type used in the mesh.
- Using the Inflation option, you can create layers of cells near the walls with a controlled growth rate. What is the purpose of using inflation layers near the walls?

### Deliverables:

- Provide a step-by-step report outlining the meshing strategy you used to generate the computational mesh.
- Include a visual representation of the generated mesh.
- Select a mesh quality metric and evaluate the mesh based on that metric. Justify your choice of the metric and explain its relevance to the specific characteristics of your mesh.

# 3. Flow Simulation

In this section, we will set up the simulation in Ansys Fluent. The following steps must be taken to generate the required deliverables in this section.

- Under Setup/General, make sure the Solver Type is set to Pressure-Based and Time is set to steady.
- Under Setup/Models, set the Viscous model to Realizable k-ε with Standard Wall Functions. What is the role of the Realizable k-ε turbulence model, and why are wall functions needed?
- Under Setup/Models, turn the Energy equation on.
- Under Setup/Material/Fluid, add water-liquid as a new material (double click on air/Fluent data bases/search for the desired material/copy)
- Under Setup/Cell Zone Condition, set the fluid material to water-liquid.
- Under Setup/Boundary conditions:
  - Neckar_in (Type: Velocity inlet): Velocity Magnitude 4 m/s, Temperature 300 K.
  - Rhine_in (Type: Velocity inlet): Velocity Magnitude 2 m/s, Temperature 290 K.
  - Rhine_out (Type: Pressure outlet) Gauge pressure 0 Pa.
- Under Setup/Reference Values, compute from Rhine_in.
- Under Solution/Methods, set Pseudo Time Method to Off.
- Under Solution/Controls, in the Limits section change the Maximum Turb. Viscosity ratio to 1e10. Why might it be necessary to increase this limit?
- Under Solution/Monitors/Residuals, change all the residuals except continuity to 1e-6.
- Under Solution/Initialization, select Standard Initialization, Compute from Rhine_in.

**Deliverables:**

After setting up the simulation with the above-mentioned conditions, run the simulation for up to 1000 iterations (the solution should converge before reaching 1000 iterations) and report the following:

- A contour of the Turbulent Kinetic Energy (TKE)
- Compare the temperature distribution plots at two different cross-sections: one immediately downstream of the junction and one before the outlet. (*Hint: you can either perform this analysis in CFD post, same as tutorial, or directly in Ansys Fluent by defining custom lines on the geometry, through the Line/Rake feature under Results/Surfaces*). How can you quantify the quality of temperature mixing from this analysis?
- Area-weighted average temperature at Rhine_out (Reports/Surface Integrals).

Please submit the deliverables as a single PDF file per group by **November 13th at 23h59** to: mohammad.jamshidmofid@epfl.ch

Best of luck with the project!